

INTERNAL

NOAA climate anomalies vs energy and grain positioning.

Author: Cerberus Research

Generated: 2026-08-05 08:21 UTC

N	EDGE	HIT-RATE
348	+6.84%	96.3%
CLASSIFICATION	internal	
AUTHOR	Cerberus Research	
RUN ID	20260805_092118	
GIT COMMIT	aa4576bd	
DURATION	1.9s	
DATASETS	noaa/noaa_climate_features, cot/cot_curated	
PARAMETER · LOOKBACK_YEARS	10	
PARAMETER · HORIZON	21	
PARAMETER · COMMODITY	USO	
KPI	Value	
N	348	

EDGE
+6.84%

HIT-RATE
96.3%

KEY FINDINGS

· Climate anomaly series: noaacl_tavg

CONTENTS

- 01 Executive Summary
- 02 Results & Interpretation
- 03 Key Findings
- 04 Data Sources
- 05 Objectives
- 06 Methodology
- 07 Charts
- 08 Statistical Results
- 09 Limitations

Generated by the Cerberus studies framework · schema 2

Data Sources

- noaa/noaa_climate_features
- cot/cot_curated

Key Findings

- Climate anomaly series: noaacl_tavg

Executive Summary

NOTE

Links NOAA climate anomaly features to commodity forward returns and futures positioning.

Results & Interpretation

- Across 348 events, the forward return averaged +6.84%. Returns were positive 96.3% of the time.
- Across 347 events, the forward return averaged +6.86%. Returns were positive 95.1% of the time.
- Across 347 events, the forward return averaged +6.10%. Returns were positive 89.3% of the time.
- Across 347 events, the forward return averaged +6.47%. Returns were positive 93.4% of the time.
- Across 343 events, the forward return averaged +6.22%. Returns were positive 91.3% of the time.

Objectives

NOTE

Evaluate whether the studied signal produces a measurable, repeatable edge.

Quantify the reliability of the observed patterns across the sample.

Methodology

NOTE

Analysis is run on the configured datasets with the parameters listed in the Study Configuration block. Results are aggregated into summary statistics, tables, and visual one-pagers for review.

Charts

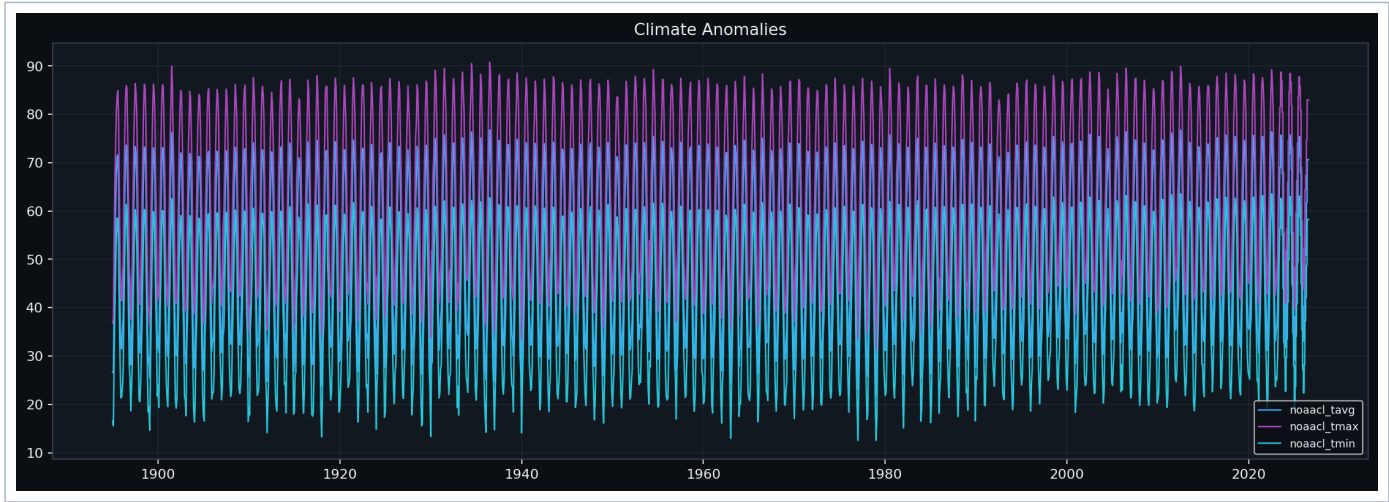


Figure 1. Climate anomalies
Climate anomalies

Statistical Results

noaac...	noacl...	noaac...	noaac...	noacl_tavg_r...	noacl_tmax_r...	noacl_tmin_r...	noacl_prcp_r...	noacl_tavg_same_...	noacl_prcp_same_...
67.51	79.34	55.69	3.57	72.14	84.49	59	3.14	1.72	2.39
53.71	64.36	43.03	3.43	64.90	76.54	53	3.33	-0.30	1.98
40.01	50.14	29.86	2.71	53.74	64.61	42	3.24	-0.94	0.81
35.51	44.78	26.26	3.25	43.08	53.09	33	3.13	0.99	1.57
32.56	42.30	22.84	2.56	36.03	45.74	26	2.84	0.66	0.53
31.84	41.70	21.96	3.28	33.30	42.93	23	3.03	-0.66	2.58
40.28	51.37	29.19	2.24	34.89	45.12	24	2.69	-0.53	-0.54
52.83	64.85	40.82	3.21	41.65	52.64	30	2.91	0.86	1.49
59.49	70.84	48.11	4.47	50.87	62.35	39	3.31	-0.54	2.95
68.70	81.18	56.23	3.37	60.34	72.29	48	3.68	-0.01	0.92
74.59	87.08	62.11	2.76	67.59	79.70	55	3.53	0.67	-0.07
73.72	86.32	61.11	2.78	72.34	84.86	59	2.97	1.22	0.39
68.41	80.58	56.25	2.45	72.24	84.66	59	2.66	2.28	-0.13
52.03	64.18	39.88	3.19	64.72	77.03	52	2.81	-1.26	1.55
41.09	52.30	29.89	1.94	53.84	65.69	42	2.53	-0.42	-0.47
36.43	46.06	26.78	2.57	43.18	54.18	32	2.57	1.32	0.33
35.53	45.03	26.02	2.76	37.68	47.80	27	2.42	1.61	0.89
36.16	47.12	25.21	2.45	36.04	46.07	26	2.59	0.67	0.72
46.08	56.71	35.44	2.86	39.26	49.62	28	2.69	1.42	0.75
50.86	63.61	38.12	2.52	44.37	55.81	32	2.61	-0.20	-0.05
60.89	73.38	48.40	3.09	52.61	64.57	40	2.82	0.35	0.27

noaac...	noacl...	noaac...	noaac...	noacl_tavg_r...	noacl_tmax_r...	noacl_tmin_r...	noacl_prcp_r...	noacl_tavg_same_...	noacl_prcp_same_...
70.27	83.03	57.51	2.73	60.67	73.34	48	2.78	1.10	-0.47
75.63	88.38	62.87	2.94	68.93	81.60	56	2.92	1.50	0.42
74.68	87.82	61.54	2.42	73.53	86.41	60	2.70	1.98	-0.60
65.89	78.80	52.99	2.44	72.07	85	59	2.60	0.55	-0.15
54.28	66.96	41.59	2.18	64.95	77.86	52	2.35	0.05	-0.09
46.31	58.30	34.30	1.94	55.49	68.02	42	2.19	1.98	-0.47
35.71	46.58	24.82	2.05	45.43	57.28	33	2.06	1.04	-0.62
34.59	43.97	25.20	2.09	38.87	49.62	28	2.03	1.29	-0.37
30.61	40.82	20.41	2.02	33.64	43.79	23	2.05	-1.06	-0.25
45.52	57.87	33.15	2.51	36.91	47.55	26	2.21	1.21	0.02
51.85	64.87	38.82	2.06	42.66	54.52	30	2.20	0.32	-1.10
60.33	72.84	47.80	2.98	52.57	65.19	39	2.52	-0.00	0.05
72.57	85.59	59.56	2.97	61.58	74.43	48	2.67	2.62	0.05
75.40	87.60	63.18	3.40	69.43	82.01	56	3.12	1.33	1.65
73.96	86.27	61.61	3.14	73.98	86.49	61	3.17	1.37	1.37
67.78	81.09	54.48	2.39	72.38	84.99	59	2.98	1.80	-0.26
56.95	68.52	45.36	3.15	66.23	78.63	53	2.89	1.58	1.47
45.07	56.84	33.28	1.30	56.60	68.82	44	2.28	1.38	-1.52
39.34	49.95	28.74	2.40	47.12	58.44	35	2.28	2.35	0.02
31.17	42.91	19.44	1.64	38.53	49.90	27	1.78	0.19	-1.22
33.66	46.26	21.04	1.76	34.72	46.37	23	1.93	-0.12	-0.82

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noaac...	noacl...	noaac...	noaac...	noacl_tavg_r...	noacl_tmax_r...	noacl_tmin_r...	noacl_prdp_r...	noacl_tavg_same_...	noacl_prdp_same_...
44.08	56.75	31.41	2.28	36.30	48.64	23	1.89	0.72	-0.46
50.67	63.25	38.08	2.60	42.80	55.42	30	2.21	-0.32	0.12
61.92	74.46	49.37	3.19	52.22	64.82	39	2.69	1.00	0.46
70.79	83.71	57.83	2.36	61.13	73.81	48	2.72	1.37	-1.27
76.39	89.22	63.55	2.78	69.70	82.46	56	2.78	2.10	-0.03
74.59	86.92	62.24	3.08	73.92	86.62	61	2.74	1.85	1.18
68.02	81.21	54.84	1.85	73.00	85.78	60	2.57	1.93	-1.46
55.18	68.58	41.81	1.68	65.93	78.90	52	2.20	0.56	-0.92
40.96	51.91	30	2.44	54.72	67.23	42	1.99	-0.49	0.38
33.12	42.58	23.65	2.79	43.09	54.36	31	2.30	0.02	0.74
35.08	44.37	25.79	2.94	36.39	46.29	26	2.72	1.42	1.21
36.34	47.62	25.05	2.01	34.85	44.86	24	2.58	0.71	-0.24
40.53	51.42	29.64	2.88	37.32	47.80	26	2.61	-0.47	0.81
51.28	64.18	38.39	2.42	42.72	54.41	31	2.44	0.01	-0.30
62.37	75.07	49.66	2.60	51.39	63.56	39	2.63	1.27	-0.68
68.99	81.57	56.39	2.87	60.88	73.61	48	2.63	0.14	-0.16
68.99	81.57	56.39	2.87	60.88	73.61	48	2.63	0.14	-0.16
75.65	88.72	62.60	2.74	69.00	81.79	56	2.74	1.47	-0.14
75.65	88.72	62.60	2.74	69.00	81.79	56	2.74	1.47	-0.14
75.65	88.72	62.60	2.74	69.00	81.79	56	2.74	1.47	-0.14
75.65	88.72	62.60	2.74	69.00	81.79	56	2.74	1.47	-0.14

noaac...	noacl...	noaac...	noaac...	noacl_tavg_r...	noacl_tmax_r...	noacl_tmin_r...	noacl_prdp_r...	noacl_tavg_same_...	noacl_prdp_same_...
75.65	88.72	62.60	2.74	69.00	81.79	56	2.74	1.47	-0.14
74.35	86.92	61.79	2.78	73.00	85.74	60	2.80	1.62	0.37
74.35	86.92	61.79	2.78	73.00	85.74	60	2.80	1.62	0.37
74.35	86.92	61.79	2.78	73.00	85.74	60	2.80	1.62	0.37
74.35	86.92	61.79	2.78	73.00	85.74	60	2.80	1.62	0.37
74.35	86.92	61.79	2.78	73.00	85.74	60	2.80	1.62	0.37
67.75	80.46	55.04	2.13	72.58	85.37	59	2.55	1.71	-0.83
67.75	80.46	55.04	2.13	72.58	85.37	59	2.55	1.71	-0.83
67.75	80.46	55.04	2.13	72.58	85.37	59	2.55	1.71	-0.83
67.75	80.46	55.04	2.13	72.58	85.37	59	2.55	1.71	-0.83
67.75	80.46	55.04	2.13	72.58	85.37	59	2.55	1.71	-0.83
56.03	68.09	43.95	2.18	66.04	78.49	53	2.36	1.03	-0.11
56.03	68.09	43.95	2.18	66.04	78.49	53	2.36	1.03	-0.11
56.03	68.09	43.95	2.18	66.04	78.49	53	2.36	1.03	-0.11
56.03	68.09	43.95	2.18	66.04	78.49	53	2.36	1.03	-0.11
56.03	68.09	43.95	2.18	66.04	78.49	53	2.36	1.03	-0.11
56.03	68.09	43.95	2.18	66.04	78.49	53	2.36	1.03	-0.11
44.22	55.65	32.79	1.39	56.00	68.07	43	1.90	0.99	-1.36
44.22	55.65	32.79	1.39	56.00	68.07	43	1.90	0.99	-1.36
44.22	55.65	32.79	1.39	56.00	68.07	43	1.90	0.99	-1.36
44.22	55.65	32.79	1.39	56.00	68.07	43	1.90	0.99	-1.36

noaac...	noacl...	noaac...	noaac...	noacl_tavg_r...	noacl_tmax_r...	noacl_tmin_r...	noacl_prdp_r...	noacl_tavg_same_...	noacl_prdp_same_...
44.22	55.65	32.79	1.39	56.00	68.07	43	1.90	0.99	-1.36
39.76	49.73	29.80	2.59	46.67	57.82	35	2.05	2.43	0.36
39.76	49.73	29.80	2.59	46.67	57.82	35	2.05	2.43	0.36
39.76	49.73	29.80	2.59	46.67	57.82	35	2.05	2.43	0.36
39.76	49.73	29.80	2.59	46.67	57.82	35	2.05	2.43	0.36
39.76	49.73	29.80	2.59	46.67	57.82	35	2.05	2.43	0.36
31.86	41	22.71	3.26	38.61	48.79	28	2.41	0.39	1.77
31.86	41	22.71	3.26	38.61	48.79	28	2.41	0.39	1.77
31.86	41	22.71	3.26	38.61	48.79	28	2.41	0.39	1.77
31.86	41	22.71	3.26	38.61	48.79	28	2.41	0.39	1.77
31.86	41	22.71	3.26	38.61	48.79	28	2.41	0.39	1.77
31.86	41	22.71	3.26	38.61	48.79	28	2.41	0.39	1.77
40.78	51.87	29.70	2.05	37.47	47.53	27	2.63	2.05	-0.14
40.78	51.87	29.70	2.05	37.47	47.53	27	2.63	2.05	-0.14
40.78	51.87	29.70	2.05	37.47	47.53	27	2.63	2.05	-0.14
40.78	51.87	29.70	2.05	37.47	47.53	27	2.63	2.05	-0.14
40.78	51.87	29.70	2.05	37.47	47.53	27	2.63	2.05	-0.14
44.94	56.37	33.49	2.88	39.19	49.75	28	2.73	1.00	0.81
44.94	56.37	33.49	2.88	39.19	49.75	28	2.73	1.00	0.81
44.94	56.37	33.49	2.88	39.19	49.75	28	2.73	1.00	0.81
44.94	56.37	33.49	2.88	39.19	49.75	28	2.73	1.00	0.81

noaac...	noacl...	noaac...	noaac...	noacl_tavg_r...	noacl_tmax_r...	noacl_tmin_r...	noacl_prdp_r...	noacl_tavg_same_...	noacl_prdp_same_...
44.94	56.37	33.49	2.88	39.19	49.75	28	2.73	1.00	0.81
53.60	66.02	41.18	2.82	46.44	58.09	34	2.58	1.25	0.62
53.60	66.02	41.18	2.82	46.44	58.09	34	2.58	1.25	0.62
53.60	66.02	41.18	2.82	46.44	58.09	34	2.58	1.25	0.62
53.60	66.02	41.18	2.82	46.44	58.09	34	2.58	1.25	0.62
53.60	66.02	41.18	2.82	46.44	58.09	34	2.58	1.25	0.62
53.60	66.02	41.18	2.82	46.44	58.09	34	2.58	1.25	0.62
62.26	74.64	49.86	3.64	53.60	65.68	41	3.11	1.18	1.31
62.26	74.64	49.86	3.64	53.60	65.68	41	3.11	1.18	1.31
62.26	74.64	49.86	3.64	53.60	65.68	41	3.11	1.18	1.31
62.26	74.64	49.86	3.64	53.60	65.68	41	3.11	1.18	1.31
62.26	74.64	49.86	3.64	53.60	65.68	41	3.11	1.18	1.31
71.80	84.60	59.02	2.76	62.55	75.09	50	3.07	2.03	-0.39
71.80	84.60	59.02	2.76	62.55	75.09	50	3.07	2.03	-0.39
71.80	84.60	59.02	2.76	62.55	75.09	50	3.07	2.03	-0.39
71.80	84.60	59.02	2.76	62.55	75.09	50	3.07	2.03	-0.39
71.80	84.60	59.02	2.76	62.55	75.09	50	3.07	2.03	-0.39
75.70	88.45	62.96	3.09	69.92	82.56	57	3.16	1.50	0.81
75.70	88.45	62.96	3.09	69.92	82.56	57	3.16	1.50	0.81
75.70	88.45	62.96	3.09	69.92	82.56	57	3.16	1.50	0.81
75.70	88.45	62.96	3.09	69.92	82.56	57	3.16	1.50	0.81

noaac...	noacl...	noaac...	noaac...	noacl_tavg_r...	noacl_tmax_r...	noacl_tmin_r...	noacl_prdp_r...	noacl_tavg_same_...	noacl_prdp_same_...
75.70	88.45	62.96	3.09	69.92	82.56	57	3.16	1.50	0.81
75.70	88.45	62.96	3.09	69.92	82.56	57	3.16	1.50	0.81
74.01	86.72	61.29	2.46	73.84	86.59	61	2.77	1.32	-0.50
74.01	86.72	61.29	2.46	73.84	86.59	61	2.77	1.32	-0.50
74.01	86.72	61.29	2.46	73.84	86.59	61	2.77	1.32	-0.50
74.01	86.72	61.29	2.46	73.84	86.59	61	2.77	1.32	-0.50
74.01	86.72	61.29	2.46	73.84	86.59	61	2.77	1.32	-0.50
68.61	81.84	55.36	2.30	72.77	85.67	59	2.62	2.23	-0.44
68.61	81.84	55.36	2.30	72.77	85.67	59	2.62	2.23	-0.44
68.61	81.84	55.36	2.30	72.77	85.67	59	2.62	2.23	-0.44
68.61	81.84	55.36	2.30	72.77	85.67	59	2.62	2.23	-0.44
68.61	81.84	55.36	2.30	72.77	85.67	59	2.62	2.23	-0.44
58.93	73.11	44.74	0.96	67.18	80.56	53	1.91	2.59	-2.07
58.93	73.11	44.74	0.96	67.18	80.56	53	1.91	2.59	-2.07
58.93	73.11	44.74	0.96	67.18	80.56	53	1.91	2.59	-2.07
58.93	73.11	44.74	0.96	67.18	80.56	53	1.91	2.59	-2.07
58.93	73.11	44.74	0.96	67.18	80.56	53	1.91	2.59	-2.07
45.14	55.51	34.75	3.04	57.56	70.15	44	2.10	1.39	1.36
45.14	55.51	34.75	3.04	57.56	70.15	44	2.10	1.39	1.36
45.14	55.51	34.75	3.04	57.56	70.15	44	2.10	1.39	1.36
45.14	55.51	34.75	3.04	57.56	70.15	44	2.10	1.39	1.36

noaac...	noacl...	noaac...	noaac...	noacl_tavg_r...	noacl_tmax_r...	noacl_tmin_r...	noacl_prcp_r...	noacl_tavg_same_...	noacl_prcp_same_...
45.14	55.51	34.75	3.04	57.56	70.15	44	2.10	1.39	1.36
38.16	48.31	27.99	2.38	47.41	58.98	35	2.13	1.80	-0.03
38.16	48.31	27.99	2.38	47.41	58.98	35	2.13	1.80	-0.03
38.16	48.31	27.99	2.38	47.41	58.98	35	2.13	1.80	-0.03
38.16	48.31	27.99	2.38	47.41	58.98	35	2.13	1.80	-0.03
38.16	48.31	27.99	2.38	47.41	58.98	35	2.13	1.80	-0.03
38.16	48.31	27.99	2.38	47.41	58.98	35	2.13	1.80	-0.03
29.17	39.81	18.54	1.37	37.49	47.88	27	2.26	-0.48	-1.70
29.17	39.81	18.54	1.37	37.49	47.88	27	2.26	-0.48	-1.70
29.17	39.81	18.54	1.37	37.49	47.88	27	2.26	-0.48	-1.70
29.17	39.81	18.54	1.37	37.49	47.88	27	2.26	-0.48	-1.70
29.17	39.81	18.54	1.37	37.49	47.88	27	2.26	-0.48	-1.70
34.77	45.90	23.63	2.17	34.03	44.67	23	1.97	0.17	0.14
34.77	45.90	23.63	2.17	34.03	44.67	23	1.97	0.17	0.14
34.77	45.90	23.63	2.17	34.03	44.67	23	1.97	0.17	0.14
34.77	45.90	23.63	2.17	34.03	44.67	23	1.97	0.17	0.14
34.77	45.90	23.63	2.17	34.03	44.67	23	1.97	0.17	0.14
46.85	59.61	34.09	2.41	36.93	48.44	25	1.98	1.62	-0.17
46.85	59.61	34.09	2.41	36.93	48.44	25	1.98	1.62	-0.17
46.85	59.61	34.09	2.41	36.93	48.44	25	1.98	1.62	-0.17
46.85	59.61	34.09	2.41	36.93	48.44	25	1.98	1.62	-0.17

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noaac...	noacl...	noaac...	noaac...	noacl_tavg_r...	noacl_tmax_r...	noacl_tmin_r...	noacl_prcp_r...	noacl_tavg_same_...	noacl_prcp_same_...
46.85	59.61	34.09	2.41	36.93	48.44	25	1.98	1.62	-0.17
53.58	66.31	40.86	2.86	45.07	57.27	32	2.48	1.21	0.71
53.58	66.31	40.86	2.86	45.07	57.27	32	2.48	1.21	0.71
53.58	66.31	40.86	2.86	45.07	57.27	32	2.48	1.21	0.71
53.58	66.31	40.86	2.86	45.07	57.27	32	2.48	1.21	0.71
53.58	66.31	40.86	2.86	45.07	57.27	32	2.48	1.21	0.71
61.72	74.07	49.37	3.68	54.05	66.66	41	2.98	0.83	1.39
61.72	74.07	49.37	3.68	54.05	66.66	41	2.98	0.83	1.39
61.72	74.07	49.37	3.68	54.05	66.66	41	2.98	0.83	1.39
61.72	74.07	49.37	3.68	54.05	66.66	41	2.98	0.83	1.39
61.72	74.07	49.37	3.68	54.05	66.66	41	2.98	0.83	1.39
71.20	83.73	58.68	3.31	62.17	74.70	49	3.28	1.58	0.80
71.20	83.73	58.68	3.31	62.17	74.70	49	3.28	1.58	0.80
71.20	83.73	58.68	3.31	62.17	74.70	49	3.28	1.58	0.80
71.20	83.73	58.68	3.31	62.17	74.70	49	3.28	1.58	0.80
71.20	83.73	58.68	3.31	62.17	74.70	49	3.28	1.58	0.80
75.42	87.76	63.07	3.07	69.45	81.85	57	3.35	1.25	0.78
75.42	87.76	63.07	3.07	69.45	81.85	57	3.35	1.25	0.78
75.42	87.76	63.07	3.07	69.45	81.85	57	3.35	1.25	0.78
75.42	87.76	63.07	3.07	69.45	81.85	57	3.35	1.25	0.78
75.42	87.76	63.07	3.07	69.45	81.85	57	3.35	1.25	0.78

noaac...	noacl...	noaac...	noaac...	noacl_tavg_r...	noacl_tmax_r...	noacl_tmin_r...	noacl_prcp_r...	noacl_tavg_same_...	noacl_prcp_same_...
73.38	85.87	60.89	2.33	73.33	85.79	60	2.90	0.81	-0.85
73.38	85.87	60.89	2.33	73.33	85.79	60	2.90	0.81	-0.85
73.38	85.87	60.89	2.33	73.33	85.79	60	2.90	0.81	-0.85
73.38	85.87	60.89	2.33	73.33	85.79	60	2.90	0.81	-0.85
73.38	85.87	60.89	2.33	73.33	85.79	60	2.90	0.81	-0.85
67.95	80.83	55.02	1.94	72.25	84.82	59	2.45	1.76	-1.24
67.95	80.83	55.02	1.94	72.25	84.82	59	2.45	1.76	-1.24
67.95	80.83	55.02	1.94	72.25	84.82	59	2.45	1.76	-1.24
67.95	80.83	55.02	1.94	72.25	84.82	59	2.45	1.76	-1.24
67.95	80.83	55.02	1.94	72.25	84.82	59	2.45	1.76	-1.24
67.95	80.83	55.02	1.94	72.25	84.82	59	2.45	1.76	-1.24
56.97	68.99	44.92	2.33	66.10	78.56	53	2.20	1.47	0.14
56.97	68.99	44.92	2.33	66.10	78.56	53	2.20	1.47	0.14
56.97	68.99	44.92	2.33	66.10	78.56	53	2.20	1.47	0.14
56.97	68.99	44.92	2.33	66.10	78.56	53	2.20	1.47	0.14
56.97	68.99	44.92	2.33	66.10	78.56	53	2.20	1.47	0.14
46.72	57.97	35.46	1.73	57.21	69.26	45	2	2.06	-0.82
46.72	57.97	35.46	1.73	57.21	69.26	45	2	2.06	-0.82
46.72	57.97	35.46	1.73	57.21	69.26	45	2	2.06	-0.82
46.72	57.97	35.46	1.73	57.21	69.26	45	2	2.06	-0.82
46.72	57.97	35.46	1.73	57.21	69.26	45	2	2.06	-0.82

noaac...	noaaci...	noaac...	noaac...	noaaci_tavg_r...	noaaci_tmax_r...	noaaci_tmin_r...	noaaci_prdp_r...	noaaci_tavg_same_...	noaaci_prdp_same_...
37.87	48.54	27.19	2.05	47.19	58.50	35	2.04	1.65	-0.64
37.87	48.54	27.19	2.05	47.19	58.50	35	2.04	1.65	-0.64
37.87	48.54	27.19	2.05	47.19	58.50	35	2.04	1.65	-0.64
37.87	48.54	27.19	2.05	47.19	58.50	35	2.04	1.65	-0.64
37.87	48.54	27.19	2.05	47.19	58.50	35	2.04	1.65	-0.64
37.87	48.54	27.19	2.05	47.19	58.50	35	2.04	1.65	-0.64
33.15	43.93	22.37	1.56	39.25	50.15	28	1.78	0.80	-1.33
33.15	43.93	22.37	1.56	39.25	50.15	28	1.78	0.80	-1.33
33.15	43.93	22.37	1.56	39.25	50.15	28	1.78	0.80	-1.33
33.15	43.93	22.37	1.56	39.25	50.15	28	1.78	0.80	-1.33
33.15	43.93	22.37	1.56	39.25	50.15	28	1.78	0.80	-1.33
40.30	52.36	28.24	1.40	37.11	48.28	25	1.67	1.88	-1.62
40.30	52.36	28.24	1.40	37.11	48.28	25	1.67	1.88	-1.62
40.30	52.36	28.24	1.40	37.11	48.28	25	1.67	1.88	-1.62
40.30	52.36	28.24	1.40	37.11	48.28	25	1.67	1.88	-1.62
40.30	52.36	28.24	1.40	37.11	48.28	25	1.67	1.88	-1.62
50.86	64.38	37.33	1.85	41.44	53.56	29	1.60	2.85	-1.34
50.86	64.38	37.33	1.85	41.44	53.56	29	1.60	2.85	-1.34
50.86	64.38	37.33	1.85	41.44	53.56	29	1.60	2.85	-1.34
50.86	64.38	37.33	1.85	41.44	53.56	29	1.60	2.85	-1.34
50.86	64.38	37.33	1.85	41.44	53.56	29	1.60	2.85	-1.34

noaac...	noacl...	noaac...	noaac...	noacl_tavg_r...	noacl_tmax_r...	noacl_tmin_r...	noacl_prcp_r...	noacl_tavg_same_...	noacl_prcp_same_...
50.86	64.38	37.33	1.85	41.44	53.56	29	1.60	2.85	-1.34
54.77	67.46	42.10	2.72	48.64	61.40	35	1.99	1.82	0.37
54.77	67.46	42.10	2.72	48.64	61.40	35	1.99	1.82	0.37
54.77	67.46	42.10	2.72	48.64	61.40	35	1.99	1.82	0.37
54.77	67.46	42.10	2.72	48.64	61.40	35	1.99	1.82	0.37
54.77	67.46	42.10	2.72	48.64	61.40	35	1.99	1.82	0.37
61.72	74.68	48.74	2.89	55.78	68.84	42	2.49	0.81	-0.13
61.72	74.68	48.74	2.89	55.78	68.84	42	2.49	0.81	-0.13
61.72	74.68	48.74	2.89	55.78	68.84	42	2.49	0.81	-0.13
61.72	74.68	48.74	2.89	55.78	68.84	42	2.49	0.81	-0.13
61.72	74.68	48.74	2.89	55.78	68.84	42	2.49	0.81	-0.13
70.63	82.98	58.26	3.23	62.37	75.04	49	2.95	1.18	0.63
70.63	82.98	58.26	3.23	62.37	75.04	49	2.95	1.18	0.63
70.63	82.98	58.26	3.23	62.37	75.04	49	2.95	1.18	0.63
70.63	82.98	58.26	3.23	62.37	75.04	49	2.95	1.18	0.63
70.63	82.98	58.26	3.23	62.37	75.04	49	2.95	1.18	0.63
70.63	82.98	58.26	3.23	62.37	75.04	49	2.95	1.18	0.63
70.63	82.98	58.26	3.23	62.37	75.04	49	2.95	1.18	0.63
70.63	82.98	58.26	3.23	62.37	75.04	49	2.95	1.18	0.63
70.63	82.98	58.26	3.23	62.37	75.04	49	2.95	1.18	0.63
70.63	82.98	58.26	3.23	62.37	75.04	49	2.95	1.18	0.63
70.63	82.98	58.26	3.23	62.37	75.04	49	2.95	1.18	0.63

Table 1. Climate anomaly data

cotcur_long_p...	cotcur_short_p...	cotcur_open_i...	cotcur_trader...	cotcur_trader...	cotcur_net_p...	cotcur_net_positio...	cotcur_wow...	cotcur_z_...
89,911.84	77,101.58	437,983	40.42	27.42	12,810	0	13,316.32	0.82
85,335.50	79,741.30	423,319	37.30	28.30	5,594	0	-7,394.16	0.66
89,134.30	86,403.85	428,901	39.45	29.05	2,730	0	-2,863.75	0.61
89,129.70	91,971.10	436,505	40.15	29	-2,841	0	-4,855.68	0.49
89,129.70	91,971.10	436,505	40.15	29	-2,841	0	-4,855.68	0.49
93,781.68	101,655.26	459,932	40.53	31.42	-7,873	0	-4,134.58	0.31
86,059.55	91,877.27	409,587	35.73	29.59	-5,817	0	143	0.34
84,545.73	90,741.91	405,458	36.05	28.77	-6,196	0	-571.55	0.26
89,249.36	95,791.36	410,932	36.09	29.23	-6,542	0	-345.82	0.17
89,249.36	95,791.36	410,932	36.09	29.23	-6,542	0	-345.82	0.17
87,848.26	90,311.17	403,737	36.74	28.70	-2,462	0	3,880.35	0.32
90,149.17	90,153.91	442,760	35.65	27.78	-4	0	2,458.17	0.35
86,895.55	87,370.30	414,700	38.45	29.70	-474	0	-1,429.30	0.01
92,231.16	95,084.53	436,453	40.05	30.21	-2,853	0	-1,462.16	-0.18
92,231.16	95,084.53	436,453	40.05	30.21	-2,853	0	-1,462.16	-0.18
86,729.14	81,365.33	400,262	38	28.90	5,363	0	7,680	-0.01
90,236	81,496.50	420,147	38.30	29.40	8,739	0	2,852.35	-0.16
91,024.29	77,350.76	408,304	36.76	30	13,673	0	5,593.19	-0.22
90,386.40	79,935.65	424,714	38.10	29.95	10,450	0	-3,422.90	-0.43
78,617.91	73,279.18	385,272	35.18	27.27	5,338	0	-3,635.45	-0.42

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cotcur_long_p...	cotcur_short_p...	cotcur_open_i...	cotcur_trader...	cotcur_trader...	cotcur_net_p...	cotcur_net_positio...	cotcur_wow...	cotcur_z_...
78,617.91	73,279.18	385,272	35.18	27.27	5,338	0	-3,635.45	-0.42
86,295.65	78,972.10	420,034	36.75	28.40	7,323	0	1,426.50	-0.22
85,735.86	77,262.86	407,975	37.29	27.48	8,473	0	1,494.29	-0.12
89,225.73	76,749.64	397,234	36.77	26.73	12,476	0	3,778.32	0.09
95,530.95	79,794.33	422,494	37.86	27	15,736	0	2,822.62	0.14
95,530.95	79,794.33	422,494	37.86	27	15,736	0	2,822.62	0.14
88,409.10	77,135.76	416,736	36.52	27.05	11,273	0	-4,376.67	0.05
83,266.39	77,894.70	403,082	33.17	27	5,371	0	-5,118.22	-0.05
90,702.82	85,077.41	446,016	34.86	27.27	5,625	0	-348.27	-0.11
88,328.79	83,792.84	422,978	39.47	29.37	4,535	0	-3,599.84	-0.05
88,328.79	83,792.84	422,978	39.47	29.37	4,535	0	-3,599.84	-0.05
88,671.40	79,985.10	411,875	38.90	27	8,686	0	5,460.25	0.06
92,558.30	85,275.95	422,186	40.05	27.50	7,282	0	-1,403.95	-0.01
97,870.95	85,761.76	414,414	39.57	27.43	12,109	0	5,717.14	0.16
96,539.14	80,164.14	409,074	39.43	26.33	16,375	0	4,265.81	0.10
97,457	81,570.25	424,448	39.20	28.40	15,886	0	-1,099.40	0.16
97,457	81,570.25	424,448	39.20	28.40	15,886	0	-1,099.40	0.16
96,477.79	76,607.21	435,963	38.53	27.68	19,870	0	3,227.63	0.20
86,079.33	71,048.86	400,442	36.38	25.81	15,030	0	-3,217.71	0.14
89,308.86	77,692.10	406,456	36.81	26.24	11,616	0	-3,413.71	0.08
90,025.77	75,449.64	398,125	35.77	25.05	14,576	0	3,845.23	0.25

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cotcur_long_p...	cotcur_short_p...	cotcur_open_i...	cotcur_trader...	cotcur_trader...	cotcur_net_p...	cotcur_net_positio...	cotcur_wow...	cotcur_z_...
90,025.77	75,449.64	398,125	35.77	25.05	14,576	0	3,845.23	0.25
96,902.33	74,620.33	417,078	37	25	22,282	0	6,323.86	0.37
96,461.43	76,029.57	424,997	35.81	25.38	20,431	0	-1,850.14	0.34
89,926.59	75,960.27	437,146	36.45	23.50	13,966	0	-4,880.18	0.39
99,978.70	68,709.25	415,046	39.05	24.45	31,269	0	15,196.40	0.35
100,570.05	67,930.35	420,275	38.60	24.15	32,639	0	1,370.25	0.33
96,506.29	69,329.90	408,913	36.38	23.19	27,176	0	-3,976.33	0.18
95,828.68	68,528.95	398,085	36.36	23.59	27,299	0	2,068.41	0.12
101,486.95	76,182.62	419,627	37.71	25.10	25,304	0	-3,350.67	0.04
102,873.95	75,254.33	423,697	37.43	25.33	27,619	0	2,315.29	0.03
102,873.95	75,254.33	423,697	37.43	25.33	27,619	0	2,315.29	0.03
102,862.19	69,955.86	421,692	35.76	24.90	32,906	0	5,286.71	0.19
102,095.76	77,471.10	434,635	36.43	26.62	24,624	0	-8,281.67	0.13
94,650.59	74,319.95	421,421	33.55	25.86	20,330	0	-3,121.95	0.02
90,211.73	78,473.23	426,024	33.82	26.82	11,738	0	-8,592.14	0.02
90,211.73	78,473.23	426,024	33.82	26.82	11,738	0	-8,592.14	0.02
95,524.95	83,587.52	447,393	35.52	28.38	11,937	0	-624.90	-0.01
95,803.23	80,705.82	446,998	34.36	27.68	15,097	0	3,955.41	0.15
94,949.95	79,113.18	462,272	35.14	26.18	15,836	0	739.36	0.08
79,665.86	75,045.57	400,214	32.86	25.33	4,620	0	-12,886.10	-0.24
78,491.71	74,475.81	400,643	32.86	24.76	4,015	0	-604.38	-0.33

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cotcur_long_p...	cotcur_short_p...	cotcur_open_i...	cotcur_trader...	cotcur_trader...	cotcur_net_p...	cotcur_net_positio...	cotcur_wow...	cotcur_z_...
78,491.71	74,475.81	400,643	32.86	24.76	4,015	0	-604.38	-0.33
83,202.20	79,976.80	423,903	33.85	27.25	3,225	0	-1,407.70	-0.27
78,616.67	74,288.19	408,025	32.57	26.86	4,328	0	1,653.24	-0.45
83,352	79,914.90	407,740	32.52	27.24	3,437	0	-891.38	-0.33
77,696.73	74,171.55	396,018	31.91	27.50	3,525	0	1,118.18	-0.43
77,696.73	74,171.55	396,018	31.91	27.50	3,525	0	1,118.18	-0.43
92,470.80	82,626.15	436,834	34.95	28.65	9,844	0	6,266.80	-0.28
94,195.71	83,602.38	423,032	33.14	28.86	10,593	0	1,187.43	-0.20
90,632.27	82,350.36	409,041	32.14	28.32	8,281	0	-2,073.82	-0.29
90,557.50	80,760.95	415,225	33.32	27.77	9,796	0	1,514.64	-0.36
90,557.50	80,760.95	415,225	33.32	27.77	9,796	0	1,514.64	-0.36
99,026.14	79,202.50	422,377	34.77	27.91	19,823	0	10,027.09	-0.16
100,525.45	74,813.09	450,288	36.68	26.82	25,712	0	5,888.73	-0.10
98,805.23	70,489.05	467,420	36.68	25.09	28,316	0	2,603.82	0.00
87,199	68,209.52	412,699	34.57	24.95	18,989	0	-13,192.55	-0.11
89,089.05	69,473.36	402,522	33.82	25	19,615	0	2,794.23	-0.04
88,975.95	71,716.64	427,045	32.73	24.41	17,259	0	-2,356.36	-0.00
84,043.86	69,041.18	415,059	33.41	22.82	15,002	0	-2,256.64	-0.03
85,850	70,685.09	418,953	35.45	22.91	15,164	0	162.23	0.04
83,237.91	70,611.14	416,984	33.55	23.59	12,626	0	-2,538.14	-0.01
83,237.91	70,611.14	416,984	33.55	23.59	12,626	0	-2,538.14	-0.01

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cotcur_long_p...	cotcur_short_p...	cotcur_open_i...	cotcur_trader...	cotcur_trader...	cotcur_net_p...	cotcur_net_positio...	cotcur_wow...	cotcur_z_...
83,817.95	74,049.18	420,881	34.73	23.95	9,768	0	-2,858	-0.10
84,861.23	77,405.68	421,415	32.95	24	7,455	0	-2,313.23	-0.16
84,370	75,296.27	420,701	32.32	24.64	9,073	0	1,618.18	-0.10
87,606.27	72,122.73	422,856	33.23	22.73	15,483	0	6,409.82	0.10
87,606.27	72,122.73	422,856	33.23	22.73	15,483	0	6,409.82	0.10
85,272.87	70,678.22	410,654	32.91	23	14,594	0	128.52	0.13
83,851.70	73,913.35	424,679	32.87	23.78	9,938	0	-4,656.30	0.04
85,497.61	81,703.48	444,397	32.87	23.96	3,794	0	-6,144.22	-0.17
86,561.86	72,247.86	403,254	34.14	22.71	14,314	0	9,061.95	-0.01
88,070.41	66,319.86	391,263	34.05	23.50	21,750	0	8,869.09	0.18
89,091.41	73,110	396,612	34	23.50	15,981	0	-5,769.14	0.01
89,632.50	76,604.64	404,894	34.50	23.77	13,027	0	-5,769.14	0.01
88,701.96	75,915.35	391,544	32.87	24.30	12,786	0	-2,246.65	-0.05
92,583.05	82,756.09	413,994	32.95	24.36	9,826	0	-3,273.18	-0.28
92,583.05	82,756.09	413,994	32.95	24.36	9,826	0	-3,273.18	-0.28
91,332.18	81,890.23	409,065	33.68	23.50	9,441	0	-385	-0.15
88,749	84,982.65	401,902	33.87	23.83	3,766	0	-5,521.13	-0.22
93,278.95	88,690.73	422,255	35.05	24.91	4,588	0	692.36	-0.27
85,663.38	80,684.04	394,498	32.54	23.12	4,979	0	641.29	-0.17
85,663.38	80,684.04	394,498	32.54	23.12	4,979	0	641.29	-0.17
91,030.65	81,080.78	413,692	34.70	26.30	9,949	0	4,704.96	-0.07

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cotcur_long_p...	cotcur_short_p...	cotcur_open_i...	cotcur_trader...	cotcur_trader...	cotcur_net_p...	cotcur_net_positio...	cotcur_wow...	cotcur_z_...
93,505.83	83,905.35	430,132	35.52	25.04	9,600	0	-349.39	-0.01
88,713.71	81,273.62	442,813	34.29	24.79	7,440	0	-1,713.33	-0.04
85,195.30	73,856.65	395,940	34.65	23.30	11,338	0	3,368.96	0.23
87,713.45	74,448.18	412,853	36.27	24.09	13,265	0	1,427.77	0.07
87,713.45	74,448.18	412,853	36.27	24.09	13,265	0	1,427.77	0.07
85,438.13	74,751	400,968	35.17	24.04	10,687	0	-2,017.22	0.04
81,352.74	78,245.91	400,970	33.87	23.70	3,106	0	-7,580.30	-0.14
81,143.58	74,803.54	382,664	33.04	23.79	6,340	0	3,560.21	-0.17
82,934.62	76,276.75	389,678	33.88	23.67	6,657	0	317.83	-0.09
82,934.62	76,276.75	389,678	33.88	23.67	6,657	0	317.83	-0.09
83,127	80,233.58	395,412	32.96	23.88	2,893	0	-3,764.46	-0.22
87,480.09	84,611.43	417,150	33.30	23.61	2,868	0	102.78	-0.30
88,702.30	86,625.13	426,928	34.83	24.09	2,077	0	-791.48	-0.35
86,017.04	85,489.88	415,094	33.67	23	527	0	-1,706.25	-0.37
86,017.04	85,489.88	415,094	33.67	23	527	0	-1,706.25	-0.37
88,911.78	90,347.09	434,333	34.52	22.43	-1,435	0	-2,367.17	-0.32
94,292.39	92,542.09	460,112	35.30	22.65	1,750	0	3,185.61	-0.11
98,250.13	91,203.70	500,679	38.35	21.70	7,046	0	5,296.13	0.07
91,238.76	76,768.95	424,120	38.52	21.76	14,469	0	8,109.10	0.12
87,274.73	74,002.86	410,270	36.91	20.73	13,271	0	-268.76	0.10
87,274.73	74,002.86	410,270	36.91	20.73	13,271	0	-268.76	0.10

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cotcur_long_p...	cotcur_short_p...	cotcur_open_i...	cotcur_trader...	cotcur_trader...	cotcur_net_p...	cotcur_net_positio...	cotcur_wow...	cotcur_z_...
89,129.55	73,968.55	414,132	37	21.86	15,161	0	529.45	0.36
87,593.04	77,677	397,151	36.65	22.35	9,916	0	-4,314.43	0.16
86,589.87	76,873.04	396,606	36.74	22.78	9,716	0	-199.22	0.00
89,287.22	76,805.35	398,588	38.17	22.30	12,481	0	2,765.04	0.27
89,287.22	76,805.35	398,588	38.17	22.30	12,481	0	2,765.04	0.27
94,740.05	83,316.27	418,339	38.23	23.41	11,423	0	-1,795.50	0.21
92,177.39	77,247.13	406,387	36.57	23.65	14,930	0	4,384.96	0.22
92,413.05	83,972.41	431,434	35.82	25.14	8,440	0	-7,182.77	0.07
87,052.96	83,456.17	418,278	35.04	23.35	3,596	0	-4,463	-0.07
87,052.96	83,456.17	418,278	35.04	23.35	3,596	0	-4,463	-0.07
88,019.35	88,732.83	432,251	34.48	24.57	-713	0	-4,310.26	-0.27
79,107.50	82,162.38	422,061	30.46	24.58	-3,054	0	-2,691.57	-0.40
77,386.04	80,444.71	435,620	30.67	25.17	-3,058	0	-3.79	-0.46
75,192.78	75,296.52	389,697	30.52	25	-103	0	2,986.43	-0.46
77,050.08	72,446.38	383,387	30.25	22.62	4,603	0	4,800.38	-0.14
77,050.08	72,446.38	383,387	30.25	22.62	4,603	0	4,800.38	-0.14
78,016.13	77,817.04	397,873	30.70	23.74	199	0	-4,197.87	-0.36
79,699.83	82,229.96	391,941	31.21	24.08	-2,530	0	-3,110.88	-0.29
82,265.83	86,493.62	397,428	32.33	24.96	-4,227	0	-1,697.67	-0.41
81,850.40	86,928.12	388,639	30.64	24.40	-5,077	0	-1,211.88	-0.62
81,850.40	86,928.12	388,639	30.64	24.40	-5,077	0	-1,211.88	-0.62

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cotcur_long_p...	cotcur_short_p...	cotcur_open_i...	cotcur_trader...	cotcur_trader...	cotcur_net_p...	cotcur_net_positio...	cotcur_wow...	cotcur_z_...
84,122.96	85,643.36	392,709	31.48	23.08	-1,520	0	3,557.32	-0.45
84,985.48	91,692.60	402,098	31.80	24.04	-6,707	0	-5,186.72	-0.62
87,693.88	95,884	407,004	31.12	24.88	-8,190	0	-1,483	-0.71
85,631.16	96,376.04	411,949	31.20	24.52	-10,744	0	-2,554.76	-0.60
85,631.16	96,376.04	411,949	31.20	24.52	-10,744	0	-2,554.76	-0.60
90,845.50	107,689.92	438,623	32.92	25.71	-16,844	0	-5,696.41	-0.57
90,992.58	107,943.21	450,171	31.58	26.79	-16,950	0	-106.21	-0.47
88,629.32	102,336.68	477,428	30.56	24.92	-13,707	0	2,540.88	-0.23
75,847.75	82,164.12	397,041	30.04	25.50	-6,316	0	8,575.33	-0.03
75,724.21	86,294.58	403,953	29.21	25.33	-10,570	0	-4,254	-0.06
78,356.48	88,606.52	415,098	31.74	25.35	-10,250	0	580.83	-0.07
77,375.67	88,137.79	398,996	30	24.42	-10,762	0	-26,886.95	-0.07
75,073.32	85,762.88	386,112	29.16	24.80	-10,689	0	-1,100	-0.06
74,486.85	85,099.23	382,366	28.50	23.58	-10,612	0	-900	0.04

Table 2. COT positioning data

regime	count	avg_fwd_ret_...	median_fwd_ret_...	hit_rate_pct
Q1	348	+ 6.84%	+ 7.34%	+ 96.26%
Q2	347	+ 6.86%	+ 7.34%	+ 95.10%
Q3	347	+ 6.10%	+ 7.34%	+ 89.34%
Q4	347	+ 6.47%	+ 7.34%	+ 93.37%
Q5	343	+ 6.22%	+ 7.34%	+ 91.25%

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Table 3. Forward returns by climate regime

Limitations

NOTE

Results are research-grade, not investment advice. The sample is limited to available data, and measured edges may not persist out-of-sample.

Appendix Tables

climate

Unnamed: 0	noaa...	noaa...	noaa...	noaa...	noacl_tavg...	noacl_tma...	noacl_tmin...	noacl_prctp...	noacl_tavg_sa...	noacl_prctp_sa...
2018-09-01 00:00:00+00:00	67.51	79.34	55.69	3.57	72.14	84.49	59	3.14	1.72	2.39
2018-10-01 00:00:00+00:00	53.71	64.36	43.03	3.43	64.90	76.54	53	3.33	-0.30	1.98
2018-11-01 00:00:00+00:00	40.01	50.14	29.86	2.71	53.74	64.61	42	3.24	-0.94	0.81
2018-12-01 00:00:00+00:00	35.51	44.78	26.26	3.25	43.08	53.09	33	3.13	0.99	1.57
2019-01-01 00:00:00+00:00	32.56	42.30	22.84	2.56	36.03	45.74	26	2.84	0.66	0.53
2019-02-01 00:00:00+00:00	31.84	41.70	21.96	3.28	33.30	42.93	23	3.03	-0.66	2.58
2019-03-01 00:00:00+00:00	40.28	51.37	29.19	2.24	34.89	45.12	24	2.69	-0.53	-0.54
2019-04-01 00:00:00+00:00	52.83	64.85	40.82	3.21	41.65	52.64	30	2.91	0.86	1.49
2019-05-01 00:00:00+00:00	59.49	70.84	48.11	4.47	50.87	62.35	39	3.31	-0.54	2.95
2019-06-01 00:00:00+00:00	68.70	81.18	56.23	3.37	60.34	72.29	48	3.68	-0.01	0.92
2019-07-01 00:00:00+00:00	74.59	87.08	62.11	2.76	67.59	79.70	55	3.53	0.67	-0.07
2019-08-01 00:00:00+00:00	73.72	86.32	61.11	2.78	72.34	84.86	59	2.97	1.22	0.39
2019-09-01 00:00:00+00:00	68.41	80.58	56.25	2.45	72.24	84.66	59	2.66	2.28	-0.13
2019-10-01 00:00:00+00:00	52.03	64.18	39.88	3.19	64.72	77.03	52	2.81	-1.26	1.55

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Unnamed: 0	noaa...	noaa...	noaa...	noaa...	noaacl_tavg...	noaacl_tma...	noaacl_tmin...	noaacl_prcp...	noaacl_tavg_sa...	noaacl_prcp_sa...
2019-11-01 00:00:00+00:00	41.09	52.30	29.89	1.94	53.84	65.69	42	2.53	-0.42	-0.47

cot_positioning

Unnamed: 0	cotcur_long...	cotcur_shor...	cotcur_open...	cotcur_trad...	cotcur_trad...	cotcur_net...	cotcur_net_posi...	cotcur_wo...	cotcur_...
2024-01-09 00:00:00+00:00	89,911.84	77,101.58	437,983	40.42	27.42	12,810	0	13,316.32	0.82
2024-01-16 00:00:00+00:00	85,335.50	79,741.30	423,319	37.30	28.30	5,594	0	-7,394.16	0.66
2024-01-23 00:00:00+00:00	89,134.30	86,403.85	428,901	39.45	29.05	2,730	0	-2,863.75	0.61
2024-01-30 00:00:00+00:00	89,129.70	91,971.10	436,505	40.15	29	-2,841	0	-4,855.68	0.49
2024-02-01 00:00:00+00:00	89,129.70	91,971.10	436,505	40.15	29	-2,841	0	-4,855.68	0.49
2024-02-06 00:00:00+00:00	93,781.68	101,655.26	459,932	40.53	31.42	-7,873	0	-4,134.58	0.31
2024-02-13 00:00:00+00:00	86,059.55	91,877.27	409,587	35.73	29.59	-5,817	0	143	0.34
2024-02-20 00:00:00+00:00	84,545.73	90,741.91	405,458	36.05	28.77	-6,196	0	-571.55	0.26
2024-02-27 00:00:00+00:00	89,249.36	95,791.36	410,932	36.09	29.23	-6,542	0	-345.82	0.17
2024-03-01 00:00:00+00:00	89,249.36	95,791.36	410,932	36.09	29.23	-6,542	0	-345.82	0.17
2024-03-05 00:00:00+00:00	87,848.26	90,311.17	403,737	36.74	28.70	-2,462	0	3,880.35	0.32

Unnamed: 0	cotcur_long...	cotcur_shor...	cotcur_open...	cotcur_trad...	cotcur_trad...	cotcur_net...	cotcur_net_posi...	cotcur_wo...	cotcur_...
2024-03-12 00:00:00+00:00	90,149.17	90,153.91	442,760	35.65	27.78	-4	0	2,458.17	0.35
2024-03-19 00:00:00+00:00	86,895.55	87,370.30	414,700	38.45	29.70	-474	0	-1,429.30	0.01
2024-03-26 00:00:00+00:00	92,231.16	95,084.53	436,453	40.05	30.21	-2,853	0	-1,462.16	-0.18
2024-04-01 00:00:00+00:00	92,231.16	95,084.53	436,453	40.05	30.21	-2,853	0	-1,462.16	-0.18

regime_fwd_returns

regime	count	avg_fwd_ret_...	median_fwd_ret_...	hit_rate_pct
Q1	348	+ 6.84%	+ 7.34%	+ 96.26%
Q2	347	+ 6.86%	+ 7.34%	+ 95.10%
Q3	347	+ 6.10%	+ 7.34%	+ 89.34%
Q4	347	+ 6.47%	+ 7.34%	+ 93.37%
Q5	343	+ 6.22%	+ 7.34%	+ 91.25%

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Statistics Summary

Key	Value
study	climate_commodity_impact
category	alternative_data
status	success
parameters	{'lookback_years': 10, 'horizon': 21, 'commodity': 'USO'}
datasets	['noaa/noaa_climate_features', 'cot/cot_curated']
rows	5080
git_commit	aa4576bdd6426fdeffd63347e0768e52b6c94632
executed_at	2026-08-05T08:21:18.875142+00:00

Glossary

Term	Definition
n / count	Number of observations or events in the sample.
edge	Average excess return versus the unconditional baseline.
hit-rate	Share of events with a positive forward return.
p-value	Probability of observing the result under the null of no edge.
*** / ** / *	Significance at 99.9% / 99% / 95% confidence.
α (alpha)	Intercept or active return versus benchmark.
β (beta)	Sensitivity to a reference factor or market.
σ (sigma)	Volatility / standard deviation of returns.

Document Control

Field	Value
Report schema	2
Framework version	0.1.0
Classification	internal
Generated	2026-08-05 08:21 UTC
Author	Cerberus Research
Study slug	climate_commodity_impact
Run ID	20260805_092118
Git commit	aa4576bd
Datasets	noaa/noaa_climate_features, cot/cot_curated

Field	Value
Parameters hash	45386cdca674
Parameter · lookback_years	10
Parameter · horizon	21
Parameter · commodity	USO
Data fingerprint	156f03406dfd
Duration	1.9s

List of Tables

- Table 1: Climate anomaly data
- Table 2: COT positioning data
- Table 3: Forward returns by climate regime